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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21BT501 / 20BT501 – REACTION ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** question from **Unit – II & Unit – III** each.

2) Linear graph may be provided.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Write a short note on the role of reaction engineering in bioprocess industries. | 5 | L*1 | 1 | 2 |
| b) Differentiate between elementary and non-elementary reaction. | 5 | L1 | 1 | 2 |
| c) Derive a relationship between equilibrium constant and temperature. Calculate the standard free energy of the reaction at 317 K and 391 K and average value of heat of reaction over this temperature range, if equilibrium is attained at 31mol % (317K) and 43mol% (391K).
Assume: Activities are equal to the mole fractions. | 10 | L3 | 2 | 1 |
| 2. a) The following table shows how the concentration of reactant A varied with time in a particular experiment. | | | | |

Time, min	0	18	31	55	79	157
C_A (mol/L) $\times 10^4$	2.77	2.32	2.05	1.59	1.26	0.58

(i) Find the progress of reaction at time 10, 50, 100 and 150min.

(ii) Find the value of rate constant and order of reaction.

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|--|----|----|---|---|
| b) What is the criterion of chemical reaction equilibria? Explain. | 10 | L2 | 1 | 2 |
|--|----|----|---|---|

Unit – II

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|---|---|----|---|---|
| 3. a) At 1100K n-nonane thermally cracks 20times as rapidly as at 1000K. Estimate the activation energy for this decomposition. | 4 | L3 | 3 | 2 |
| b) Decomposition of an acetone dicarboxylic acid is a first order. Following are the data for the same. | | | | |

T, K	273	293	313	333
$k \times 10^5$	2.46	47.5	576	5480

Find out the energy of activation for this reaction graphically.

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|--|----|----|---|---|
| c) Explain different approach to determine temperature dependent term of rate equation. | 6 | L3 | 3 | 2 |
| 4. a) With suitable example explain initial rate method of analyzing the kinetic data. | 10 | L2 | 3 | 2 |
| b) Calculate the first order rate constant for the disappearance of A as per the gas phase reaction $A \rightarrow 1.6R$ if the volume of the reaction mixture, starting with pure A increases by 50% in 4min. the total pressure of the system remains at 1.2atm and the temperature is 25°C. | 10 | L3 | 3 | 3 |
| 5. a) Using integral method, derive an expression to analyzing the kinetic data for the reaction of the type $2A \rightarrow R$. | 6 | L3 | 3 | 2 |

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- b) Sucrose is hydrolyzed at room temperature by the catalytic action of the enzyme sucrase as $\text{sucrose} \xrightarrow{\text{sucrase}} \text{product}$. Following kinetic data are obtained in a batch reactor by starting with a sucrose concentration of $C_{A0}=1.0\text{mmol/L}$ and enzyme concentration $C_{E0}=0.01\text{mmol/L}$.

$C_A, \text{mmol/L}$	0.68	0.38	0.16	0.04	0.006
Time, h	2	4	6	8	10

Obtain a kinetic equation and its constants for this reaction.

14 L3 3 3

Unit – III

6. a) Derive an equation to determine the conversion for first order reaction carried out in batch reactor.
b) In detail explain the significance of E and F curve in determining degree of non-ideality in reactor.

10 L3 4 2

10 L2 5 3

7. a) The dye degradation from a paint shop of an automotive industry is carried out in a MFR. The holding time and extent of degradation is given below.

Holding Time, Sec	0.423	5.1	13.5	44	192
Extent of degradation	0.22	0.63	0.75	0.88	0.96

If initial concentration of dye in the effluent is 0.002mol/L , determine the kinetics of degradation.

10 L3 4 2

- b) With suitable mathematical expression prove that the mean residence time is equal to space time ($\bar{t} = \tau$).

10 L3 5 3

8. a) Derive the performance equation for plug flow reactor (PFR) in terms of reactant conversion.

10 L3 4 2

- b) The degree of non-ideality in a newly designed tank fermenter was estimated by measuring RTD. A known amount of tracer was injected by pulse mode. The effluent concentration was measured as a function of time. The data obtained are as follows:

t, min	0	2	4	6	8	10	12	14
C, g/m ³	0	5	10	6	3	1.5	0.6	0

Construct E curve and determine the fraction of material leaving the vessel between time 4 and 10min.

10 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
